

A Derivation of the Hsieh–Klenow (HK)(Hsieh and Klenow, 2009) Framework

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0. Notation and Structure

- Industries (sectors) are indexed by $s = 1, \dots, S$.
- Varieties/firms within industry s are indexed by $i = 1, \dots, M_s$.
- Final-good Cobb–Douglas weights: $\theta_s > 0$, with

$$\sum_{s=1}^S \theta_s = 1.$$

- Within-industry CES elasticity of substitution: $\sigma > 1$.
- Capital share in industry s : $d_s \in (0, 1)$.
- Distortions (wedges):
 - output/revenue wedge τ_{Ysi} (the firm keeps fraction $(1 - \tau_{Ysi})$ of revenue),
 - capital wedge τ_{Ksi} (the effective rental rate is $(1 + \tau_{Ksi})R$).
- Factor prices: wage w , undistorted rental rate R .

We derive, step by step, the chain:

Final good $\rightarrow$ Industry CES $\rightarrow$ Firm (distorted) $\rightarrow$ MRP/TFPR $\rightarrow$ Industry TFP.

1. Final-Good Market: Cost Minimization and the Final Price Index

1.1 Final-good aggregator and cost

The final good Y is produced from industry outputs $\{Y_s\}_{s=1}^S$ via Cobb–Douglas:

$$Y = \prod_{s=1}^S Y_s^{\theta_s}. \tag{1}$$

Let P_s be the price of industry output Y_s . A representative final-good producer chooses $\{Y_s\}$ to minimize total cost:

$$\min_{\{Y_s\}_{s=1}^S} \sum_{s=1}^S P_s Y_s \quad (2)$$

$$\text{s.t. } Y = \prod_{s=1}^S Y_s^{\theta_s}. \quad (3)$$

1.2 Lagrangian and first-order condition

The Lagrangian is

$$\mathcal{L} = \sum_{s=1}^S P_s Y_s + \lambda \left(Y - \prod_{s=1}^S Y_s^{\theta_s} \right). \quad (4)$$

We need $\partial/\partial Y_s$ of the product term. Since

$$\prod_{u=1}^S Y_u^{\theta_u} = Y,$$

its derivative w.r.t. Y_s is

$$\frac{\partial}{\partial Y_s} \left(\prod_{u=1}^S Y_u^{\theta_u} \right) = \theta_s \left(\prod_{u=1}^S Y_u^{\theta_u} \right) \frac{1}{Y_s} = \theta_s \frac{Y}{Y_s}. \quad (5)$$

Thus the FOC is

$$\frac{\partial \mathcal{L}}{\partial Y_s} = P_s - \lambda \theta_s \frac{Y}{Y_s} = 0 \quad (6)$$

$$\implies P_s = \lambda \theta_s \frac{Y}{Y_s} \quad (7)$$

$$\implies Y_s = \lambda \theta_s \frac{Y}{P_s}. \quad (1)$$

1.3 Solving for λ and defining the final price index P

Substitute (1) into the constraint:

$$Y = \prod_{s=1}^S \left(\lambda \theta_s \frac{Y}{P_s} \right)^{\theta_s} \quad (8)$$

$$= (\lambda Y)^{\sum_s \theta_s} \left(\prod_{s=1}^S \theta_s^{\theta_s} \right) \left(\prod_{s=1}^S P_s^{-\theta_s} \right). \quad (9)$$

Since $\sum_s \theta_s = 1$, this becomes

$$Y = \lambda Y \left(\prod_{s=1}^S \theta_s^{\theta_s} \right) \left(\prod_{s=1}^S P_s^{-\theta_s} \right). \quad (10)$$

Divide both sides by Y :

$$1 = \lambda \left(\prod_{s=1}^S \theta_s^{\theta_s} \right) \left(\prod_{s=1}^S P_s^{-\theta_s} \right). \quad (11)$$

Rearrange:

$$\lambda = \left(\prod_{s=1}^S \theta_s^{\theta_s} \right)^{-1} \left(\prod_{s=1}^S P_s^{\theta_s} \right) \quad (12)$$

$$= \prod_{s=1}^S \left(\frac{P_s}{\theta_s} \right)^{\theta_s}. \quad (13)$$

Define the final-good price index P (unit cost) as

$$P \equiv \prod_{s=1}^S \left(\frac{P_s}{\theta_s} \right)^{\theta_s}, \quad (14)$$

so $\lambda = P$.

1.4 Expenditure shares across industries

From the FOC form $P_s = \lambda \theta_s Y / Y_s$, multiply both sides by Y_s :

$$P_s Y_s = \lambda \theta_s Y = P \theta_s Y, \quad (15)$$

hence

$$\boxed{P_s Y_s = \theta_s P Y.} \quad (16)$$

2. Industry Level: CES Aggregator, Variety Demand, and Industry Price Index

2.1 CES aggregator

Within industry s , output is a CES aggregate of varieties:

$$Y_s = \left(\sum_{i=1}^{M_s} Y_{si}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}. \quad (17)$$

Equivalently, raise both sides to power $(\sigma - 1)/\sigma$:

$$Y_s^{\frac{\sigma-1}{\sigma}} = \sum_{i=1}^{M_s} Y_{si}^{\frac{\sigma-1}{\sigma}}. \quad (2)$$

2.2 Cost minimization

Let P_{si} be the price of variety i in industry s . An industry aggregator chooses $\{Y_{si}\}$ to minimize cost for given Y_s :

$$\min_{\{Y_{si}\}} \sum_{i=1}^{M_s} P_{si} Y_{si} \quad (18)$$

$$\text{s.t. } Y_s^{\frac{\sigma-1}{\sigma}} = \sum_{i=1}^{M_s} Y_{si}^{\frac{\sigma-1}{\sigma}}. \quad (19)$$

The Lagrangian (multiplier μ_s) is

$$\mathcal{L}_s = \sum_i P_{si} Y_{si} + \mu_s \left(Y_s^{\frac{\sigma-1}{\sigma}} - \sum_i Y_{si}^{\frac{\sigma-1}{\sigma}} \right). \quad (20)$$

2.3 First-order condition and relative demand

FOC for Y_{si} :

$$\frac{\partial \mathcal{L}_s}{\partial Y_{si}} = P_{si} - \mu_s \frac{\sigma-1}{\sigma} Y_{si}^{\frac{\sigma-1}{\sigma}-1} = 0. \quad (21)$$

Since $\frac{\sigma-1}{\sigma} - 1 = -\frac{1}{\sigma}$, we have $Y_{si}^{\frac{\sigma-1}{\sigma}-1} = Y_{si}^{-1/\sigma}$, so the FOC becomes

$$P_{si} = \mu_s \frac{\sigma-1}{\sigma} Y_{si}^{-1/\sigma}. \quad (3)$$

For any i, j , divide the FOCs:

$$\frac{P_{si}}{P_{sj}} = \frac{Y_{si}^{-1/\sigma}}{Y_{sj}^{-1/\sigma}} = \left(\frac{Y_{sj}}{Y_{si}} \right)^{1/\sigma}. \quad (22)$$

Raise both sides to power σ and rearrange:

$$\left(\frac{P_{si}}{P_{sj}} \right)^\sigma = \frac{Y_{sj}}{Y_{si}} \quad (23)$$

$$\implies Y_{si} = Y_{sj} \left(\frac{P_{sj}}{P_{si}} \right)^\sigma. \quad (4)$$

2.4 Industry price index and standard variety demand

Define the CES industry price index

$$P_s \equiv \left(\sum_{i=1}^{M_s} P_{si}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}. \quad (5)$$

Then equivalently

$$P_s^{1-\sigma} = \sum_{i=1}^{M_s} P_{si}^{1-\sigma}. \quad (6)$$

With this definition, the standard CES demand takes the form

$$Y_{si} = Y_s \left(\frac{P_s}{P_{si}} \right)^\sigma. \quad (7)$$

(One can verify (7) by substituting into (2) and using (6).)

3. Firm Level: Cost Minimization, Marginal Cost, and Factor Ratios (with Capital Wedges)

3.1 Production and effective factor prices

Firm i in industry s produces

$$Y_{si} = A_{si} K_{si}^{d_s} L_{si}^{1-d_s}, \quad (24)$$

with effective rental rate

$$r_{si} \equiv (1 + \tau_{Ksi})R. \quad (25)$$

3.2 Cost minimization for given output y

For any target output $y > 0$, solve

$$\min_{K,L} \quad wL + r_{si}K \quad (26)$$

$$\text{s.t.} \quad A_{si} K^{d_s} L^{1-d_s} \geq y. \quad (27)$$

At optimum the constraint binds, so use equality. Lagrangian with multiplier ν :

$$\mathcal{C} = wL + r_{si}K + \nu \left(y - A_{si} K^{d_s} L^{1-d_s} \right). \quad (28)$$

FOCs:

$$\frac{\partial \mathcal{C}}{\partial K} = r_{si} - \nu A_{si} d_s K^{d_s-1} L^{1-d_s} = 0 \quad (29)$$

$$\implies r_{si} = \nu A_{si} d_s K^{d_s-1} L^{1-d_s}, \quad (8)$$

and

$$\frac{\partial \mathcal{C}}{\partial L} = w - \nu A_{si} (1 - d_s) K^{d_s} L^{-d_s} = 0 \quad (30)$$

$$\implies w = \nu A_{si} (1 - d_s) K^{d_s} L^{-d_s}. \quad (9)$$

3.3 Optimal factor ratio K/L

Divide (8) by (9):

$$\frac{r_{si}}{w} = \frac{d_s}{1 - d_s} \cdot \frac{K^{d_s-1} L^{1-d_s}}{K^{d_s} L^{-d_s}} = \frac{d_s}{1 - d_s} \cdot \frac{L}{K}. \quad (31)$$

Rearrange:

$$\frac{K}{L} = \frac{d_s}{1 - d_s} \cdot \frac{w}{r_{si}} = \frac{d_s}{1 - d_s} \cdot \frac{w}{(1 + \tau_{Ksi})R}. \quad (10)$$

3.4 Deriving marginal cost MC_{si} with no skipped steps

Because the technology is constant returns to scale, the cost function is linear in y :

$$C_{si}(y) = MC_{si} y.$$

We now derive MC_{si} explicitly.

Let $\phi_{si} \equiv K/L$. Then $K = \phi_{si}L$, and the production function becomes

$$y = A_{si}(\phi_{si}L)^{d_s} L^{1-d_s} = A_{si}\phi_{si}^{d_s} L. \quad (32)$$

Thus

$$L = \frac{y}{A_{si}\phi_{si}^{d_s}}. \quad (11)$$

Total cost is

$$C = wL + r_{si}K = wL + r_{si}\phi_{si}L = (w + r_{si}\phi_{si})L. \quad (33)$$

Using (10), we have

$$r_{si}\phi_{si} = r_{si} \left(\frac{d_s}{1-d_s} \frac{w}{r_{si}} \right) = \frac{d_s}{1-d_s} w, \quad (34)$$

so

$$w + r_{si}\phi_{si} = w + \frac{d_s}{1-d_s} w = \frac{w}{1-d_s}. \quad (35)$$

Therefore

$$C = \frac{w}{1-d_s} L = \frac{w}{1-d_s} \cdot \frac{y}{A_{si}\phi_{si}^{d_s}} \quad (\text{using (11)}). \quad (36)$$

Now compute $\phi_{si}^{d_s}$:

$$\phi_{si}^{d_s} = \left(\frac{d_s}{1-d_s} \frac{w}{r_{si}} \right)^{d_s} = \frac{d_s^{d_s}}{(1-d_s)^{d_s}} \cdot \frac{w^{d_s}}{r_{si}^{d_s}}. \quad (37)$$

Hence

$$\frac{1}{\phi_{si}^{d_s}} = \frac{(1-d_s)^{d_s}}{d_s^{d_s}} \cdot \frac{r_{si}^{d_s}}{w^{d_s}}. \quad (38)$$

Substitute this into C :

$$C = \frac{w}{1-d_s} \cdot \frac{y}{A_{si}} \cdot \frac{(1-d_s)^{d_s}}{d_s^{d_s}} \cdot \frac{r_{si}^{d_s}}{w^{d_s}} \quad (39)$$

$$= \frac{y}{A_{si}} \cdot \frac{w^{1-d_s} r_{si}^{d_s}}{d_s^{d_s} (1-d_s)^{1-d_s}}. \quad (40)$$

Thus marginal cost is

$$\boxed{MC_{si} = \frac{1}{A_{si}} \cdot \frac{w^{1-d_s} ((1 + \tau_{Ksi})R)^{d_s}}{d_s^{d_s} (1-d_s)^{1-d_s}}}. \quad (12)$$

4. Firm Pricing: CES Demand + Revenue Wedge $\Rightarrow$ Optimal Price

4.1 Variety demand

From (7), firm i faces

$$Y_{si} = Y_s \left(\frac{P_s}{P_{si}} \right)^\sigma. \quad (13)$$

4.2 Profit with distortions

The firm keeps fraction $(1 - \tau_{Ysi})$ of revenue, and pays factor cost according to MC_{si} :

$$\pi_{si} = (1 - \tau_{Ysi})P_{si}Y_{si} - wL_{si} - r_{si}K_{si} \quad (41)$$

$$= (1 - \tau_{Ysi})P_{si}Y_{si} - MC_{si}Y_{si}. \quad (14)$$

Using (13), rewrite quantity as a function of price:

$$Y_{si} = Y_s P_s^\sigma P_{si}^{-\sigma}. \quad (15)$$

4.3 Solve for optimal P_{si}

Substitute (15) into (14):

$$\pi_{si} = (1 - \tau_{Ysi})P_{si} \cdot Y_s P_s^\sigma P_{si}^{-\sigma} - MC_{si} \cdot Y_s P_s^\sigma P_{si}^{-\sigma} \quad (42)$$

$$= Y_s P_s^\sigma [(1 - \tau_{Ysi})P_{si}^{1-\sigma} - MC_{si}P_{si}^{-\sigma}]. \quad (43)$$

Differentiate w.r.t. P_{si} :

$$\frac{d}{dP} P^{1-\sigma} = (1 - \sigma)P^{-\sigma}, \quad \frac{d}{dP} P^{-\sigma} = -\sigma P^{-\sigma-1}.$$

Thus

$$\frac{d\pi_{si}}{dP_{si}} = Y_s P_s^\sigma [(1 - \tau_{Ysi})(1 - \sigma)P_{si}^{-\sigma} - MC_{si}(-\sigma)P_{si}^{-\sigma-1}] = 0 \quad (44)$$

$$\iff (1 - \tau_{Ysi})(1 - \sigma)P_{si}^{-\sigma} + MC_{si}\sigma P_{si}^{-\sigma-1} = 0 \quad (\text{divide by } Y_s P_s^\sigma > 0) \quad (45)$$

$$\iff (1 - \tau_{Ysi})(1 - \sigma)P_{si} + MC_{si}\sigma = 0 \quad (\text{multiply by } P_{si}^{\sigma+1} > 0) \quad (46)$$

$$\iff (\sigma - 1)(1 - \tau_{Ysi})P_{si} = \sigma MC_{si} \quad (\text{since } 1 - \sigma = -(\sigma - 1)). \quad (47)$$

Hence the optimal price is

$$P_{si} = \frac{\sigma}{\sigma - 1} \cdot \frac{MC_{si}}{1 - \tau_{Ysi}}. \quad (16)$$

4.4 Plugging in MC_{si} for an explicit pricing formula

Substitute (12) into (16). Define the common industry constant

$$\Xi_s \equiv \frac{\sigma}{\sigma - 1} \left(\frac{w}{1 - d_s} \right)^{1-d_s} \left(\frac{R}{d_s} \right)^{d_s}.$$

Then one obtains

$$P_{si} = \frac{\sigma}{\sigma - 1} \left(\frac{w}{1 - d_s} \right)^{1-d_s} \left(\frac{R}{d_s} \right)^{d_s} \cdot \frac{(1 + \tau_{Ksi})^{d_s}}{(1 - \tau_{Ysi})A_{si}}. \quad (17)$$

5. Output and Factor Demands

5.1 Output expression

From demand (13),

$$Y_{si} = Y_s \left(\frac{P_s}{P_{si}} \right)^\sigma = Y_s P_s^\sigma P_{si}^{-\sigma}. \quad (18)$$

Using (17) and the definition of Ξ_s , we can write

$$P_{si} = \Xi_s \cdot \frac{(1 + \tau_{Ksi})^{d_s}}{(1 - \tau_{Ysi})A_{si}}.$$

Thus

$$P_{si}^{-\sigma} = \Xi_s^{-\sigma} \left(\frac{(1 - \tau_{Ysi})A_{si}}{(1 + \tau_{Ksi})^{d_s}} \right)^\sigma.$$

Plug into (18):

$$\boxed{Y_{si} = Y_s P_s^\sigma \Xi_s^{-\sigma} \left(\frac{(1 - \tau_{Ysi})A_{si}}{(1 + \tau_{Ksi})^{d_s}} \right)^\sigma}. \quad (19)$$

5.2 Factor demands (two fully equivalent derivations)

Method A: use the optimal factor ratio. Let $\phi_{si} = K_{si}/L_{si}$ from (10). Then

$$Y_{si} = A_{si}(\phi_{si}L_{si})^{d_s}L_{si}^{1-d_s} = A_{si}\phi_{si}^{d_s}L_{si}.$$

Hence

$$L_{si} = \frac{Y_{si}}{A_{si}\phi_{si}^{d_s}} = \frac{Y_{si}}{A_{si}} \left(\frac{1 - d_s}{d_s} \frac{(1 + \tau_{Ksi})R}{w} \right)^{d_s}, \quad (20)$$

and

$$K_{si} = \phi_{si}L_{si} = \frac{Y_{si}}{A_{si}} \left(\frac{1 - d_s}{d_s} \frac{(1 + \tau_{Ksi})R}{w} \right)^{d_s-1}. \quad (21)$$

Method B: use Cobb–Douglas cost shares. Under cost minimization with CRS Cobb–Douglas,

$$wL_{si} = (1 - d_s)MC_{si}Y_{si}, \quad r_{si}K_{si} = d_sMC_{si}Y_{si},$$

so

$$L_{si} = \frac{(1 - d_s)MC_{si}Y_{si}}{w}, \quad K_{si} = \frac{d_sMC_{si}Y_{si}}{r_{si}}, \quad r_{si} = (1 + \tau_{Ksi})R,$$

which is algebraically identical to (20)–(21) after substituting (12).

6. Marginal Revenue Products (MRP) and Wedges

6.1 Express price as a function of quantity

From CES demand (13),

$$\frac{Y_{si}}{Y_s} = \left(\frac{P_s}{P_{si}} \right)^\sigma \implies P_{si} = P_s \left(\frac{Y_s}{Y_{si}} \right)^{1/\sigma} = P_s Y_s^{1/\sigma} Y_{si}^{-1/\sigma}. \quad (22)$$

Nominal revenue (before applying $(1 - \tau_{Y_{si}})$) is

$$R_{si} \equiv P_{si} Y_{si} = P_s Y_s^{1/\sigma} Y_{si}^{-1/\sigma} \cdot Y_{si} = P_s Y_s^{1/\sigma} Y_{si}^{1-1/\sigma}. \quad (23)$$

6.2 Marginal revenue with respect to output

Differentiate (23) w.r.t. Y_{si} :

$$\frac{\partial R_{si}}{\partial Y_{si}} = P_s Y_s^{1/\sigma} \left(1 - \frac{1}{\sigma} \right) Y_{si}^{-1/\sigma} \quad (48)$$

$$= \frac{\sigma - 1}{\sigma} \cdot P_s Y_s^{1/\sigma} Y_{si}^{-1/\sigma}. \quad (49)$$

By (22), the last term equals P_{si} , hence

$$\boxed{\frac{\partial R_{si}}{\partial Y_{si}} = \frac{\sigma - 1}{\sigma} P_{si}}. \quad (24)$$

6.3 MRP of labor and capital via the chain rule

Since $Y_{si} = A_{si} K_{si}^{d_s} L_{si}^{1-d_s}$,

$$\frac{\partial Y_{si}}{\partial L_{si}} = (1 - d_s) \frac{Y_{si}}{L_{si}}, \quad \frac{\partial Y_{si}}{\partial K_{si}} = d_s \frac{Y_{si}}{K_{si}}.$$

Using the chain rule and (24),

$$MRPL_{si} \equiv \frac{\partial R_{si}}{\partial L_{si}} = \frac{\partial R_{si}}{\partial Y_{si}} \cdot \frac{\partial Y_{si}}{\partial L_{si}} = \left(\frac{\sigma - 1}{\sigma} P_{si} \right) \cdot (1 - d_s) \frac{Y_{si}}{L_{si}} \quad (50)$$

$$= (1 - d_s) \frac{\sigma - 1}{\sigma} \frac{P_{si} Y_{si}}{L_{si}}. \quad (25)$$

Similarly,

$$MRPK_{si} \equiv \frac{\partial R_{si}}{\partial K_{si}} = \left(\frac{\sigma - 1}{\sigma} P_{si} \right) \cdot d_s \frac{Y_{si}}{K_{si}} = d_s \frac{\sigma - 1}{\sigma} \frac{P_{si} Y_{si}}{K_{si}}. \quad (26)$$

6.4 Using profit FOCs to link MRPs to wedges

Profit is

$$\pi_{si} = (1 - \tau_{Y_{si}}) R_{si} - w L_{si} - r_{si} K_{si}, \quad r_{si} = (1 + \tau_{K_{si}}) R.$$

FOC w.r.t. labor:

$$\frac{\partial \pi_{si}}{\partial L_{si}} = (1 - \tau_{Y_{si}}) \frac{\partial R_{si}}{\partial L_{si}} - w = (1 - \tau_{Y_{si}}) MRPL_{si} - w = 0 \quad (51)$$

$$\Rightarrow \boxed{MRPL_{si} = \frac{w}{1 - \tau_{Y_{si}}}}. \quad (27)$$

FOC w.r.t. capital:

$$\frac{\partial \pi_{si}}{\partial K_{si}} = (1 - \tau_{Y_{si}}) MRPK_{si} - r_{si} = 0 \quad (52)$$

$$\Rightarrow \boxed{MRPK_{si} = \frac{(1 + \tau_{K_{si}})R}{1 - \tau_{Y_{si}}}}. \quad (28)$$

7. TFPR and Industry TFP: From Definitions to the Final Log Expression

7.1 TFPR definition and equivalences

Define firm-level revenue productivity (TFPR) as

$$TFPR_{si} \equiv \frac{P_{si} Y_{si}}{K_{si}^{d_s} L_{si}^{1-d_s}}. \quad (53)$$

Since $Y_{si} = A_{si} K_{si}^{d_s} L_{si}^{1-d_s}$, we get

$$TFPR_{si} = \frac{P_{si} \cdot A_{si} K_{si}^{d_s} L_{si}^{1-d_s}}{K_{si}^{d_s} L_{si}^{1-d_s}} = \boxed{P_{si} A_{si}}. \quad (29)$$

Substitute the optimal price (17) into (29):

$$\boxed{TFPR_{si} = \frac{\sigma}{\sigma - 1} \left(\frac{R}{d_s} \right)^{d_s} \left(\frac{w}{1 - d_s} \right)^{1-d_s} \cdot \frac{(1 + \tau_{K_{si}})^{d_s}}{1 - \tau_{Y_{si}}}}. \quad (30)$$

(Under these assumptions, $TFPR_{si}$ depends only on wedges, not on A_{si} .)

7.2 Industry TFP and industry TFPR

Define aggregate factor inputs:

$$K_s \equiv \sum_{i=1}^{M_s} K_{si}, \quad L_s \equiv \sum_{i=1}^{M_s} L_{si}.$$

Define physical TFP in industry s :

$$\boxed{TFP_s \equiv \frac{Y_s}{K_s^{d_s} L_s^{1-d_s}}}. \quad (31)$$

Define industry TFPR:

$$\boxed{TFPR_s \equiv \frac{P_s Y_s}{K_s^{d_s} L_s^{1-d_s}}}. \quad (32)$$

From (31) and (32),

$$TFPR_s = P_s \cdot TFP_s \quad (54)$$

$$\implies \boxed{TFP_s = \frac{TFPR_s}{P_s}}. \quad (33)$$

7.3 Writing $1/P_s$ as a function of variety prices

From the CES price index definition (5),

$$P_s = \left(\sum_i P_{si}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}.$$

Taking reciprocals yields

$$\frac{1}{P_s} = \left(\sum_i P_{si}^{1-\sigma} \right)^{\frac{1}{\sigma-1}}. \quad (34)$$

Substitute (34) into (33):

$$TFP_s = TFPR_s \left(\sum_i P_{si}^{1-\sigma} \right)^{\frac{1}{\sigma-1}}. \quad (35)$$

7.4 Eliminating P_{si} using $TFPR_{si} = P_{si}A_{si}$

From (29), $P_{si} = TFPR_{si}/A_{si}$. Then

$$P_{si}^{1-\sigma} = \left(\frac{TFPR_{si}}{A_{si}} \right)^{1-\sigma} = TFPR_{si}^{1-\sigma} \cdot A_{si}^{\sigma-1}. \quad (36)$$

Plug (36) into (35):

$$TFP_s = TFPR_s \left(\sum_i A_{si}^{\sigma-1} TFPR_{si}^{1-\sigma} \right)^{\frac{1}{\sigma-1}}. \quad (37)$$

7.5 Bringing $TFPR_s$ inside the sum (pure algebra, no extra assumptions)

For each i ,

$$\left(A_{si} \cdot \frac{TFPR_s}{TFPR_{si}} \right)^{\sigma-1} = A_{si}^{\sigma-1} \cdot TFPR_s^{\sigma-1} \cdot TFPR_{si}^{1-\sigma}. \quad (55)$$

Multiply both sides by $TFPR_s^{1-\sigma}$:

$$TFPR_s^{1-\sigma} \left(A_{si} \cdot \frac{TFPR_s}{TFPR_{si}} \right)^{\sigma-1} = A_{si}^{\sigma-1} TFPR_{si}^{1-\sigma}. \quad (56)$$

Summing over i :

$$\sum_i A_{si}^{\sigma-1} TFPR_{si}^{1-\sigma} = TFPR_s^{1-\sigma} \sum_i \left(A_{si} \cdot \frac{TFPR_s}{TFPR_{si}} \right)^{\sigma-1}. \quad (38)$$

Substitute (38) into (37):

$$TFP_s = TFP R_s \left[TFP R_s^{1-\sigma} \sum_i \left(A_{si} \cdot \frac{TFP R_s}{TFP R_{si}} \right)^{\sigma-1} \right]^{\frac{1}{\sigma-1}} \quad (57)$$

$$= TFP R_s \cdot (TFP R_s^{1-\sigma})^{\frac{1}{\sigma-1}} \cdot \left[\sum_i \left(A_{si} \cdot \frac{TFP R_s}{TFP R_{si}} \right)^{\sigma-1} \right]^{\frac{1}{\sigma-1}}. \quad (58)$$

Now,

$$(TFP R_s^{1-\sigma})^{\frac{1}{\sigma-1}} = TFP R_s^{\frac{1-\sigma}{\sigma-1}} = TFP R_s^{-1}.$$

Hence $TFP R_s \cdot TFP R_s^{-1} = 1$, giving the final industry-TFP formula:

$$TFP_s = \left[\sum_{i=1}^{M_s} \left(A_{si} \cdot \frac{TFP R_s}{TFP R_{si}} \right)^{\sigma-1} \right]^{\frac{1}{\sigma-1}}. \quad (39)$$

7.6 Log form

Taking logs of (39):

$$\log TFP_s = \frac{1}{\sigma-1} \log \left[\sum_{i=1}^{M_s} \left(A_{si} \cdot \frac{TFP R_s}{TFP R_{si}} \right)^{\sigma-1} \right]. \quad (40)$$

8. Summary: The Key Link-by-Link Chain (No Gaps)

1. Final-good cost minimization $\Rightarrow$ expenditure shares and final price index:

$$P_s Y_s = \theta_s P Y, \quad P = \prod_{s=1}^S \left(\frac{P_s}{\theta_s} \right)^{\theta_s}.$$

2. Industry CES cost minimization $\Rightarrow$ variety demand and industry price index:

$$Y_{si} = Y_s \left(\frac{P_s}{P_{si}} \right)^{\sigma}, \quad P_s = \left(\sum_i P_{si}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}.$$

3. Firm cost minimization $\Rightarrow$ factor ratio and marginal cost:

$$\frac{K_{si}}{L_{si}} = \frac{d_s}{1-d_s} \frac{w}{(1+\tau_{Ksi})R}, \quad MC_{si} = \frac{1}{A_{si}} \frac{w^{1-d_s} ((1+\tau_{Ksi})R)^{d_s}}{d_s^{d_s} (1-d_s)^{1-d_s}}.$$

4. Pricing under CES demand with a revenue wedge $\Rightarrow$ markup over marginal cost:

$$P_{si} = \frac{\sigma}{\sigma-1} \cdot \frac{MC_{si}}{1-\tau_{Ysi}}.$$

5. TFPR definition and substitution:

$$TFP R_{si} = P_{si} A_{si} = \frac{\sigma}{\sigma-1} \left(\frac{R}{d_s} \right)^{d_s} \left(\frac{w}{1-d_s} \right)^{1-d_s} \cdot \frac{(1+\tau_{Ksi})^{d_s}}{1-\tau_{Ysi}}.$$

6. Industry TFP as $TFP R_s/P_s$ and eliminate P_{si} via $TFP R_{si} = P_{si} A_{si}$ to obtain the final HK industry-TFP formula (39) and its log form (40).

References

Hsieh, Chang-Tai, and Peter J. Klenow. “Misallocation and Manufacturing TFP in China and India.” *The Quarterly Journal of Economics* 124, no. 4 (November 2009): 1403–1448. <https://doi.org/10.1162/qjec.2009.124.4.1403>

Other Notes

This derivation also draws on *TFP Measurement: Online Course Notes* (personal learning notes), accessed January 2026.